Adam Coscia

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EDUCATION

Ph.D. in Human-Centered Computing — Atlanta, GA
Georgia Institute of Technology — May 2026
Dissertation: Detecting and Mitigating Pedagogical Risks in Large Language Models with Visual Analytics
Committee: Dr. Alex Endert, Dr. Duen Horng (Polo) Chau, Dr. Cindy Bearfield-Xiong, Dr. Yalong Yang, Dr. Scott Crossley

B.S. in Physics — Hoboken, NJ
Stevens Institute of Technology — May 2020
Minors: Mathematics, Computer Science, *GPA:* 3.98/4.00, *Achievements:* Pinnacle Scholars, First in Class (Physics)

EMPLOYMENT

Stevens Institute of Technology — Hoboken, NJ
Assistant Professor (Tenure-Track) ● School of Computing — 2026—present
Building visual analytics, human-centered artificial intelligence, and data science tools to solve pressing human-centered, data-driven problems.

Georgia Institute of Technology — Atlanta, GA
Graduate Research Assistant ● School of Interactive Computing ● Advisor: Alex Endert — 2020—2026
Member of the Visual Analytics Lab developing interactive visual analytics tools that help people make sense of data by combining information visualization, machine learning, and human-computer interaction.

AWARDS and HONORS

VAST Challenge @ IEEE VIS Best Paper, Honorable Mention — 2026
For *"**AgentK**: Investigation of Multi-Agent LLM Dialogue via Semantic Bundling of Knowledge Graphs"*

Sigma Xi Scientific Research Honor Society — 2025
Inducted as a lifetime member

CRIDC Best Poster Award — 2025
For *"**VisPile**: Incorporating Knowledge Graphs and Large Language Models into Visual Text Analysis Tools"*

Georgia Tech Foley Scholar Award Finalist — 2024
Awarded to top graduate students shaping the future of how people interact with and value technology.

CRIDC Best Poster Award — 2023
For *"**KnowledgeVIS**: Visualizing What Language Models Have Learned"*

CRIDC Best Poster Award — 2021
For "***Lumos**: Increasing Awareness of Biases during Visual Data Analysis*"

Georgia Tech President's Fellowship — 2020
Four-year semesterly stipend award; selected upon admission from top 10% of applicant pool.

Stevens Institute of Tech Alfred M. Mayer Prize — 2020
Awarded to senior ranked first in all physics courses taken during undergraduate career.

Sigma Pi Sigma Physics Honor Society 2019
Inducted as a lifetime member by the American Institute of Physics

Stevens Distinguished Teaching Assistant 2018
Awarded to student faculty member nominated for creating outstanding classroom environment.

Stevens Presidential Scholarship 2016
Four-year, half-tuition award; selected for academic excellence in high school.

PUBLICATIONS

Journal Articles

1. **KnowledgeVIS: Interpreting Language Models by Comparing Fill-in-the-Blank Prompts**
   Adam Coscia and Alex Endert
   *IEEE Trans. on Visualization and Computer Graphics (**TVCG**), 2024*
2. **Preliminary Guidelines for Combining Data Integration and Visual Data Analysis**
   Adam Coscia, Ashley Suh, Remco Chang, and Alex Endert
   *IEEE Trans. on Visualization and Computer Graphics (**TVCG**), 2024*

Conference Proceedings

1. **Less Is More: Balancing Positive and Negative Space in Visual Concept Blending**
   Shishi Xiao, Adam Coscia, and David Laidlaw
   *ACM Symposium on User Interface Software and Technology (**UIST**), Detroit, Michigan, USA, 2026*
2. **DraftMarks: Enhancing Transparency in Human-AI Co-Writing Through Interactive Skeuomorphic Process Traces**
   Momin Siddiqui, Nikki Nasseri, Adam Coscia, Roy Pea, and Hariharan Subramonyam
   *ACM Conference on Human Factors in Computing Systems (**CHI**), Barcelona, Spain, 2026*
3. **VisPile: A Visual Analytics System for Analyzing Multiple Text Documents with Large Language Models and Knowledge Graphs**
   Adam Coscia and Alex Endert
   *The 59th Hawaii International Conference on System Sciences (**HICSS**), Lahaina, Hawai'i, USA, 2026*
4. **OnGoal: Tracking and Visualizing Conversational Goals in Multi-Turn Dialogue with Large Language Models**
   Adam Coscia, Shunan Guo, Eunyee Koh, and Alex Endert
   *ACM Symposium on User Interface Software and Technology (**UIST**), Busan, Republic of Korea, 2025*
5. **Visualizing the Provenance of Intelligent Tutor Interactions towards Responsive Pedagogy**
   Grace Guo, Aishwarya Mudgal Sunil Kumar, Adit Gupta, Adam Coscia, Chris MacLellan, and Alex Endert
   *International Conference on Advanced Visual Interfaces (**AVI**), Arenzano (Genoa), Italy, 2024*
6. **DeepSee: Multidimensional Visualizations of Seabed Ecosystems**
   Adam Coscia, Haley M. Sapers, Noah Deutsch, Malika Khurana, John S. Magyar, Sergio A. Parra, Daniel R. Utter, Rebecca L. Wipfler, David W. Caress, Eric J. Martin, Jennifer B. Paduan, Maggie Hendrie, Santiago Lombeyda, Hillary Mushkin, Alex Endert, Scott Davidoff, and Victoria J. Orphan
   *ACM Conference on Human Factors in Computing Systems (**CHI**), Honolulu, Hawai'i, USA, 2024*

7. **iScore: Visual Analytics for Interpreting How Language Models Automatically Score Summaries**
   Adam Coscia, Langdon Holmes, Wesley Morris, Joon Suh Choi, Scott Crossley, and Alex Endert
   *ACM Conference on Intelligent User Interfaces (**IUI**), Greenville, South Carolina, USA, 2024*
8. **Lumos: Increasing Awareness of Analytic Behavior during Visual Data Analysis**
   Arpit Narechania, Adam Coscia, Emily Wall, and Alex Endert
   *IEEE Trans. on Visualization and Computer Graphics (Proc. **IEEE VIS** 2021), 2022*
9. **Left, Right, and Gender: Exploring Interaction Traces to Mitigate Human Biases**
   Emily Wall, Arpit Narechania, Adam Coscia, Jamal Paden, and Alex Endert
   *IEEE Trans. on Visualization and Computer Graphics (Proc. **IEEE VIS** 2021), 2022*

Workshop Papers

1. **Targeted and Traceable Investigation of Multi-Agent LLM Dialogue via Semantic Bundling of Knowledge Graphs**
   Zeyu Hua, Adam Coscia, and Alex Endert
   *VAST Challenge @ **IEEE VIS** 2026, Boston, MA, USA, 2026*
2. **LLM Prompt Design and Evaluation for Educational Applications**
   Langdon Holmes, Adam Coscia, Scott Crossley, Joon Suh Choi, and Wesley Morris
   *Workshop on Epistemics and Decision-Making in AI-Supported Education @ The 26th International Conference on Artificial Intelligence in Education (**AIED**), Palermo, Italy, 2025*
3. **Toward a Bias-Aware Future for Mixed-Initiative Visual Analytics**
   Adam Coscia, Duen Horng (Polo) Chau, and Alex Endert
   *Workshop on TRust and EXpertise in Visual Analytics @ **IEEE VIS** 2020, virtual*

Posters

1. **Incorporating Knowledge Graphs and Large Language Models into Visual Text Analysis Tools**
   Adam Coscia, Alex Endert, Liz Richerson, Sue Mi K., Stephen S., and Tim S.
   a. *Career, Research, and Innovation Development Conference (**CRIDC**), Atlanta, GA, USA, 2025. **Best Poster Award (top 5%)***
   b. *2024 LAS Research Symposium, Raleigh, NC, USA, 2024.*
2. **iScore: Visual Analytics for Interpreting How Language Models Automatically Score Summaries**
   Adam Coscia, Langdon Holmes, Wesley Morris, Joon Suh Choi, Scott Crossley, and Alex Endert
   a. *Career, Research, and Innovation Development Conference (**CRIDC**), Atlanta, GA, USA, 2024*
   b. *C21U Symposium on Generative Futures: Revolutionizing Learning with Artificial Intelligence, Atlanta, GA, USA, 2023*
3. **KnowledgeVIS: Visualizing What Language Models Have Learned**
   Adam Coscia and Alex Endert
   *Career, Research, and Innovation Development Conference (**CRIDC**), Atlanta, GA, USA, 2023. **Best Poster Award (top 5%)***
4. **Lumos: Increasing Awareness of Biases during Visual Data Analysis**
   Arpit Narechania, Adam Coscia, Emily Wall, and Alex Endert
   *Career, Research, and Innovation Development Conference (**CRIDC**), Atlanta, GA, USA, 2021. **Best Poster Award (top 5%)***

5. **Correlating Long-Term Innovation with Success in Career Progression**
   Adam Coscia and Aron Lindberg
   a. *Business Intelligence & Analytics (BI&A) Corporate Networking Event, Hoboken, NJ, USA, 2018*
   b. *Pinnacle Scholar Summer Research Poster Session, Hoboken, NJ, USA, 2018*

INVITED TALKS and DEMOS

Collaborative Sensemaking with AI in Visual Analytics

Montclair State University	April 2026
Emory University	April 2026
Brown Visual Computing Seminar, *Brown University*	October 2025
University of Massachusetts Boston	October 2025
University of Massachusetts Amherst	October 2025
Johns Hopkins University	October 2025
MIT Lincoln Laboratory	July 2025

Detecting and Mitigating Pedagogical Risks in LLMs with Visual Analytics

Tufts University	July 2025
Institute for People and Technology (IPAT), *Georgia Institute of Technology*	April 2025
NSF AI Institute AI-ALOE Showcase, *Georgia Institute of Technology*	April 2024

KnowledgeVIS: Interpreting Language Models by Comparing Fill-in-the-Blank Prompts

Invited Journal Paper Talk, *IEEE Visualization Conference (VIS)*	October 2024

Preliminary Guidelines for Combining Data Integration and Visual Data Analysis

Invited Journal Paper Talk, *IEEE Visualization Conference (VIS)*	October 2024

Visualizing What Large Language Models Have Learned

GVU Demo Day, *Georgia Institute of Technology*	March 2023

GRANTS and FUNDING

Agentic Visual Data Analysis: Knowledge Graph Synthesis using LLMs 2026
Laboratory for Analytic Sciences, *North Carolina State University*
Co-PI: **Alex Endert**
One-year funding (full tuition + graduate stipend)

Enabling Continuous Analytic Dialogues in Visual Analytics with LLMs and Knowledge Graphs 2025
Laboratory for Analytic Sciences, *North Carolina State University*
Co-PI: **Alex Endert**
One-year funding (full tuition + graduate stipend)

Incorporating Knowledge Graphs and Large Language Models into Visual Text Analysis Tools 2024
Laboratory for Analytic Sciences, *North Carolina State University*
Co-PI: **Alex Endert**
One-year funding (full tuition + graduate stipend)

Correlating Long-Term Innovation with Success in Career Progression 2018
Pinnacle Scholar Summer Institutional Research Program, *Stevens Institute of Technology*
Co-PI: **Aron Lindberg**
Funded $5000 for research expenses and living stipend

Managing Simulations of Ion Channeling in Single Crystals 2017
International Summer Abroad Internship Program, *Katholieke Universiteit Leuven*
Co-PI: **Lino da Costa Pereira**
Funded €3000 from Katholieke Universiteit Leuven and $5000 from Stevens Institute of Technology

TEACHING

Georgia Institute of Technology Atlanta, GA
Graduate Teaching Assistant ● *CS 6730: Data Visualization Principles* Fall 2022, Fall 2025
Instructor: **Alex Endert**
Assisted professor with lectures, grading, team projects; held office hours for student support.

Stevens Institute of Technology Hoboken, NJ
Course Assistant ● *PEP 112S: Honors Electricity & Magnetism* semesterly, 2018—2020
Instructor: **Christopher Search**
Assisted professor with grading, reviews, worksheets, and testing material preparation.

Stevens Institute of Technology Hoboken, NJ
Course Assistant ● *PEP 112: Electricity & Magnetism* semesterly, 2018—2020
Instructor: **Robert Pastore**
Ran 3-hour exam reviews for an average class size of 200 students.

Stevens Institute of Technology Hoboken, NJ
Teaching Assistant ● *CS 105: Intro to Scientific Computing* semesterly, 2017—2020
Instructor: **Dimitrios Damopoulos**
Taught 15-25 students in weekly labs on CS concepts using MATLAB; helped develop course materials.

ADVISING and MENTORSHIP

Ph.D. – Research Project Advisor

Zeyu (Abigail) Hua 2026
Ph.D. in Computer Science, *Georgia Institute of Technology*
Developing new LLM-powered algorithms for analyzing and visualizing knowledge graphs.

Sujata Duwal 2025—2026
Ph.D. in Human-Centered Computing, *Georgia Institute of Technology*
Studying trust in decision-making when visualizing LLMs used in education.

Master's – Research Project Advisor

Jennifer Chandran 2025
M.S. in Human-Computer Interaction, *Georgia Institute of Technology*
Creating a new LLM-augmented pipeline that automatically transforms document sets for data analysis.

Ben Klassen 2024—2025
M.S. in Data Science & Analytics, *Georgia Institute of Technology*
Developed new data science tools that leverage LLMs to enhance intelligence in classrooms and sports analytics.

Sean Ru 2024
M.S. in Computer Science, *Georgia Institute of Technology*
Created LLM-based algorithms that embed data visualizations for improved searchability.

Bachelors – Research Project Advisor

Meng Lincoln Ngy Tang 2026
B.S. in Computer Science, *Stevens Institute of Technology*
Developing visualization techniques to safely deploy AI for trustworthy financial operations.

PROGRAMS and OUTREACH

Georgia Institute of Technology Atlanta, GA
Mentor ● *IC Buddies Mentorship Program* 2025—2026
Paired one-on-one with a graduate student, helping them adjust to grad school. Introduced them to faculty connections, discussed graduate research opportunities, and provided guidance on time/workload management.

Encouraging Women Across All Borders (EWAAB) New York, NY
Mentor ● *Beyond Mentorship Program* 2022—2023
Connected monthly one-on-one with students to discuss professional topics ranging from general professional advice, to applying for job opportunities, to discovering new fields of study.

Stevens Institute of Technology Hoboken, NJ
Mentor ● *Pinnacle Scholar Peer Advisor Program* 2017—2019
Mentored 4-6 honors freshman each academic year representing different majors. Provided guidance on internships, classes, international experiences, campus resources. Took students on excursions into Hoboken.

SERVICE and ASSOCIATIONS

Professional Societies
Member, Sigma Xi Scientific Research Honor Society (**Sigma Xi**) since 2025
Member, IEEE Computer Society (**IEEE**) since 2025
Member, Association for Computing Machinery (**ACM**) since 2023
Member, Special Interest Group on Computer-Human Interaction (**SIGCHI**) since 2023
Member, Sigma Pi Sigma (**SPS**) Physics Honor Society Lifetime, since 2019
Member, American Physical Society (**APS**) 2016—2020

Conference Program Committee
IEEE Visualization Conference (**VIS**) Short Papers 2026
IEEE Pacific Visualization Conference (**PacificVis**) Short Papers 2026—2027
VISxGenAI Workshop @ IEEE VIS 2026
VAxAutoSci Workshop @ IEEE VIS 2026

Conference Organizing Committee
IEEE Visualization Conference (**VIS**) Tech Chair 2026

Conference Student Volunteer
ACM Symposium on User Interface Software and Technology (**UIST**) 2025

Reviewer

IEEE Visualization Conference (**VIS**)	2022—2026
IEEE Pacific Visualization Conference (**PacificVis**)	2025—2027
IEEE Transactions on Visualization and Graphics (**TVCG**)	2022—2026
IEEE Computer Graphics and Applications (**CG&A**)	2026
EuroVis Conference (**EuroVis**)	2023—2026
ACM Conference on Human Factors in Computing Systems (**CHI**)	2024—2027
ACM Symposium on User Interface Software and Technology (**UIST**)	2025—2026
ACM Conference on Intelligent User Interfaces (**IUI**)	2026—2027
ACM Conference on Computer-Supported Cooperative Work & Social Computing (**CSCW**)	2025
ACM Transactions on Interactive Intelligent Systems (**TIIS**)	2024, 2026
Hawaii International Conference on System Sciences (**HICSS**)	2026
Human–Computer Interaction (**HCI**)	2026
Computer Graphics Forum (**CGF**)	2026
Computers and Graphics (**C&G**)	2025
Visual Informatics	2025

Special recognitions for outstanding reviews

- **IEEE VIS** 2023, 2026
- **ACM CHI** 2024, 2025
- **ACM IUI** 2026

RESEARCH INTERNSHIPS

MIT Lincoln Laboratory — Lexington, MA
Artificial Intelligence Research Lead ● *Artificial Intelligence Group* — Summer 2025
Advisors: **Ho Chit Siu, Rohan Paleja**
Developing novel visual analytics interfaces that enable human-in-the-loop decision-making and guidance in human-machine teaming scenarios.

Adobe Research — San Jose, CA
Machine Learning Research Lead ● *AI Experiences Lab* — Summer 2024
Advisors: **Shunan Guo, Eunyee Koh**
Built a novel chat interface with large language models (LLMs) to improve sensemaking of conversational LLM responses for everyday users using LLM-powered Adobe products.

NASA Jet Propulsion Laboratory (JPL) — Pasadena, CA
Machine Learning Research Lead ● *Human-Interfaces Group* — Summer 2023
Advisors: **Scott Davidoff, Tiago Vaquero**
Developed automated science planning capabilities for planetary missions to support multi-instrument and team-driven science using a novel demonstration paradigm.

NASA Jet Propulsion Laboratory (JPL) — Pasadena, CA
Computer Science Lead ● *Data to Discovery Program* — Summer 2021
Advisors: **Scott Davidoff, Santiago Lombeyda, Hillary Mushkin, Maggie Hendrie**
Built interactive data visualization combining linked 2D maps and 3D visualizations of taxa and geochemical values in sediment cores collected from the sea floor.

Stevens Institute of Technology — Hoboken, NJ
Undergraduate Research Assistant ● *School of Information Systems and Architecture* — 2018—2020
Advisors: **Aron Lindberg, Amir Gandomi**
Developed statistical model in Python for connecting evolutionary trajectories of digital artifacts to performance outcomes in online communities.

Katholieke Universiteit Leuven — Leuven, Belgium
Visiting Research Scholar ● *Department of Physics and Astronomy* — Summer 2017
Advisors: **Lino da Costa Pereira, Tiago Abel de Lemos Lima**
Built data visualization interface in Python for managing simulations of ion channeling in single crystals, to be used in ion beam analysis of topological materials.

INDUSTRY INTERNSHIPS

New York Life Insurance Company — New York, NY
Machine Learning / Operations Intern ● *Center for Data Science and Artificial Intelligence* — Summer 2020
Supervisor: **Paul Janis**
Engineered multiple feature extraction pipelines interfaced by Domino platform and integrated with existing Hadoop infrastructure. Produced model monitoring metric reports for stakeholders and data science team.

New York Life Insurance Company — New York, NY
Data Platform Engineering Intern ● *Center for Data Science and Analytics* — Summer 2019
Supervisor: **Paul Janis**
Built various scalable programs and data-handling procedures for multiple teams to leverage complex, low-level data lake tools with efficient, cost-effective, and easy-to-use interfaces.

PRESS

1. **This New Tool Makes AI's Role in Student Writing Visible**
   Tess Malone, Georgia Tech Office of Research, April 2026
   https://research.gatech.edu/new-tool-makes-ais-role-student-writing-visible
2. **CRIDC 2025 Awards $40,000 to competition winners**
   Brittani Hill, Georgia Tech Office of Graduate and Postdoctoral Education, February 2025
   https://grad.gatech.edu/news/cridc-2025-awards-40000-competition-winners
3. **Foley Scholars 2024 Winners and Finalists**
   Walter Rich, Georgia Tech Office of Research, November 2024
   https://research.gatech.edu/foley-scholars-2024-winners-and-finalists
4. **Visualization Tool Helps Oceanographers Predict Sediment Sample Hotspots**
   Nathan Deen, Georgia Tech School of Interactive Computing, July 2024
   https://www.cc.gatech.edu/news/visualization-tool-helps-oceanographers-predict-sediment-sample-hotspots
5. **Adam Coscia Presents Visualizations for Interpreting Large Language Models at GVU Center Research Showcase**
   NSF AI Institute for Adult Learning in Online Education (AI-ALOE), April 2023
   https://aialoe.org/coscia-presents-visualizations-for-interpreting-large-language-models/
6. **DeepSEE: A Virtual Window Under the Waves**
   Serg Parra, Schmidt Ocean Institute, November 2021
   https://schmidtocean.org/cruise-log-post/deepsee-a-virtual-window-under-the-waves/